

Trainings ▾

General Information

All maintenance procedures listed for previous intervals **must** also be performed.

For convenience, listed below are the section numbers that contain specific instructions for performing the maintenance.

The QSK19 CM500, QSK19 CM850 MCRS and QSK19 CM2150 MCRS will have the following lubricating oil filtration options.

Marine and Industrial:

- Spin-on Lubricating Oil Filters
- Spin-on Filter and Centinel™
- Eliminator™ without Centinel™
- Eliminator™ with Centinel™.

Power Generation

- The Eliminator™ and Centinel™ will **not** be offered.

Spin-on Oil Filter

Maintenance Procedures at Daily Interval (Section 3)

- Air Intake Piping - Check
- Air Cleaner Restriction² - Check
- Air Tanks and Reservoirs - Drain
- Lubricating Oil Level - Check
- Coolant Level - Check
- Air Cleaner Precleaner - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Filter, Remote-Mounted - Drain
- Fuel-Water Separator - Drain
- Fuel Filter, Remote-Mounted⁵ - Check

Maintenance Procedures at 250 Hours or 6 Months¹ (Section 4)

- Fuel Filter (Spin-On Type)⁴ - Change
- Fuel Filter (Stage 1)⁵ - Change

- Fuel Filter (Stage 2)⁵ - Change
- Lubricating Oil and Filters - Change³
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check
- Oil Analysis (see Cummins® Engines) Oil and Oil Analysis Recommendations, Bulletin 3810340 (/qs3/pubsys2/xml/en/bulletin/3810340.html) ⁷
- Coolant Filter - Change
- Air Compressor Air Cleaner Element² - Check
- Drive Belt, Alternator - Inspect for Reuse
- Drive Belt, Cooling Fan - Check
- Fan, Cooling - Inspect for Reuse
- Engine Wiring Harness - Check
- Zinc Anode - Inspect for Reuse
- Intake Manifold Pressure - Check

Maintenance Procedures at 1500 Hours or 1 Year¹ (Section 7)

- Fan Hub, Belt Driven - Check
- Crankcase Breather (External) - Clean
- Crankshaft - Check
- Batteries - Check
- Engine Mounts - Check
- Cooling System Hoses - Check
- Sea Water Pump - Check
- Engine Steam Clean - Clean
- Engine Oil Heater - Check
- Engine Coolant Heater - Check

Maintenance Procedures at 6000 Hours or 2 Years¹ (Section 8)

- Cooling System - Flush
- Fan Hub, Belt Driven - Check
- Fan Drive Idler Pulley Assembly - Check
- Water Pump - Inspect for Reuse
- Air Compressor Discharge Lines - Check
- Vibration Damper, Viscous - Check⁸

Maintenance Procedures at 10,000 Hours (Section 9)

- Fuel Injectors - Replace⁶
- Turbocharger - Check⁶

Spin-on Oil Filter with Centinel™

Maintenance Procedures at Daily Interval (Section 3)

- Air Intake Piping - Check
- Air Cleaner Restriction² - Check
- Air Tanks and Reservoirs - Drain
- Lubricating Oil Level - Check
- Coolant Level - Check
- Air Cleaner Precleaner - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Centinel™ Oil Level - Check
- Fuel Filter, Remote-Mounted - Drain
- Fuel-Water Separator - Drain
- Fuel Filter, Remote-Mounted⁵ - Check

Maintenance Procedures at 250 Hours or 6 Months¹ (Section 4)

- Fuel Filter (Spin-On Type)⁴ - Change
- Fuel Filter (Stage 1)⁵ - Change
- Fuel Filter (Stage 2)⁵ - Change
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check
- Lubricating Oil Analysis - Check⁷
- Coolant Filter - Change
- Air Compressor Air Cleaner Element² - Check
- Drive Belt, Alternator - Check
- Drive Belt, Cooling Fan - Check
- Fan, Cooling - Check
- Engine Wiring Harness - Check
- Zinc Anode - Inspect for Reuse
- Intake Manifold Pressure - Check

Maintenance Procedures at 1000 Hours or 1 Year (Section 6)

- Lubricating Oil Filter (Spin-On) - Change

Maintenance Procedures at 1500 Hours or 1 Year (Section 7)

- Fan Hub, Belt Driven - Check
- Crankcase Breather (External) - Clean
- Crankshaft - Check
- Batteries - Check
- Engine Mounts - Check
- Cooling System Hoses - Check
- Sea Water Pump - Check

- Engine Steam Cleaning - Clean
- Engine Oil Heater - Check
- Engine Coolant Heater - Check

Maintenance Procedures at 6000 Hours or 2 Year (Section 8)

- Cooling System - Flush
- Fan Hub, Belt Driven - Check
- Fan Drive Idler Pulley Assembly - Check
- Water Pump - Inspect for Reuse
- Air Compressor Discharge Lines - Check
- Vibration Damper, Viscous - Check⁸

Maintenance Procedures at 10,000 Hours (Section 9)

- Fuel Injectors - Replace⁶
- Turbocharger - Check⁶

Eliminator™ without Centinel™

Maintenance Procedures at Daily Interval (Section 3)

- Air Intake Piping - Check
- Air Cleaner Restriction² - Check
- Air Tanks and Reservoirs - Drain
- Lubricating Oil Level - Check
- Coolant Level - Check
- Air Cleaner Precleaner - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Eliminator™ Filter - Check
- Fuel Filter, Remote-Mounted - Drain
- Fuel-Water Separator - Drain
- Fuel Filter, Remote-Mounted⁵ - Check

Maintenance Procedures at 250 Hours or 6 Months¹ (Section 4)

- Fuel Filter (Spin-On Type)⁴ - Change
- Fuel Filter (Stage 1)⁵ - Change
- Fuel Filter (Stage 2)⁵ - Change
- Lubricating Oil - Change³
- Crankcase Breather Tube - Check
- Lubricating Oil Analysis - Check⁷
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check

- Coolant Filter - Change
- Air Compressor Air Cleaner Element² - Check
- Drive Belt, Alternator - Check
- Drive Belt, Cooling Fan - Check
- Fan, Cooling - Check
- Engine Wiring Harness - Check
- Zinc Anode - Inspect for Reuse
- Intake Manifold Pressure - Check

Maintenance Procedures Every 1000 Hours (Section 5)

- Eliminator™ Filter Centrifuge - Change

Maintenance Procedures at 1500 Hours or 1 Year¹ (Section 7)

- Fan Hub, Belt Driven - Check
- Crankcase Breather (External) - Clean
- Crankshaft - Check
- Batteries - Check
- Engine Mounts - Check
- Cooling System Hoses - Check
- Sea Water Pump - Check
- Engine Steam Cleaning - Clean
- Engine Oil Heater - Check
- Engine Coolant Heater - Check

Maintenance Procedures at 6000 Hours or 2 Years¹ (Section 8)

- Cooling System - Flush
- Fan Hub, Belt Driven - Check
- Fan Drive Idler Pulley Assembly - Check
- Water Pump - Inspect for Reuse
- Air Compressor Discharge Lines - Check
- Vibration Damper, Viscous - Check⁸

Maintenance Procedures at 10,000 Hours (Section 9)

- Fuel Injectors - Replace⁶
- Turbocharger - Check⁶

Eliminator™ with Centinel™

Maintenance Procedures at Daily Interval (Section 3)

- Air Intake Piping - Check
- Air Cleaner Restriction² - Check
- Air Tanks and Reservoirs - Drain
- Lubricating Oil Level - Check
- Coolant Level - Check
- Air Cleaner Precleaner - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Centinel™ Oil Level - Check
- Fuel Filter, Remote-Mounted - Drain
- Fuel-Water Separator - Drain
- Fuel Filter, Remote-Mounted⁵ - Check

Maintenance Procedures at 250 Hours or 6 Months (Section 4)

- Fuel Filter (Spin-On Type)⁴ - Change
- Fuel Filter (Stage 1)⁵ - Change
- Fuel Filter (Stage 2)⁵ - Change
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check
- Lubricating Oil Analysis- Check⁷
- Coolant Filter - Change
- Air Compressor Air Cleaner Element² - Check
- Drive Belt, Alternator - Check
- Drive Belt, Cooling Fan - Check
- Fan, Cooling - Check
- Engine Wiring Harness - Check
- Zinc Anode - Inspect for Reuse
- Intake Manifold Pressure - Check

Maintenance Procedures at 1000 Hours (Section 5)

- Eliminator™ Filter Centrifuge - Change

Maintenance Procedures at 1500 Hours or 1 Year (Section 7)

- Fan Hub, Belt Driven - Check
- Crankcase Breather (External) - Clean
- Crankshaft - Check
- Batteries - Check
- Engine Mounts - Check
- Cooling System Hoses - Check
- Sea Water Pump - Check
- Engine Steam Cleaning - Clean

- Engine Oil Heater - Check
- Engine Coolant Heater - Check

Maintenance Procedures at 6000 Hours or 2 Years (Section 8)

- Cooling System - Flush
- Fan Hub, Belt Driven - Check
- Fan Drive Idler Pulley Assembly - Check
- Water Pump - Inspect for Reuse
- Air Compressor Discharge Lines - Check
- Vibration Damper, Viscous - Check⁸

Maintenance Procedures at 10,000 Hours (Section 9)

- Fuel Injectors - Replace⁶
- Turbocharger - Check⁶

Power Generation

Maintenance Procedures at Daily Interval (Section 3)

- Air Intake Piping - Check
- Air Cleaner Restriction² - Check
- Air Tanks and Reservoirs - Drain
- Lubricating Oil Level - Check
- Coolant Level - Check
- Air Cleaner Precleaner - Check
- Fuel Filter, Remote-Mounted - Drain
- Fuel-Water Separator - Drain
- Fuel Filter, Remote-Mounted⁵ - Check

Maintenance Procedures at 250 Hours or 6 Months¹ (Section 4)

- Fuel Filter (Spin-On Type)⁴ - Change
- Fuel Filter (Stage 1)⁵ - Change
- Fuel Filter (Stage 2)⁵ - Change
- Lubricating Oil and Filters - Change³
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check
- Coolant Filter - Change
- Air Compressor Air Cleaner Element² - Check
- Drive Belt, Alternator - Check
- Drive Belt, Cooling Fan - Check
- Fan, Cooling - Check

- Engine Wiring Harness - Check
- Intake Manifold Pressure - Check

Maintenance Procedures at 1500 Hours or 1 Year¹ (Section 7)

- Fan Hub, Belt Driven - Check
- Crankcase Breather (External) - Clean
- Crankshaft - Check
- Batteries - Check
- Engine Mounts - Check
- Cooling System Hoses - Check
- Engine - Clean
- Engine Oil Heater - Check
- Engine Coolant Heater - Check

Maintenance Procedures at 6000 Hours or 2 Years¹ (Section 8)

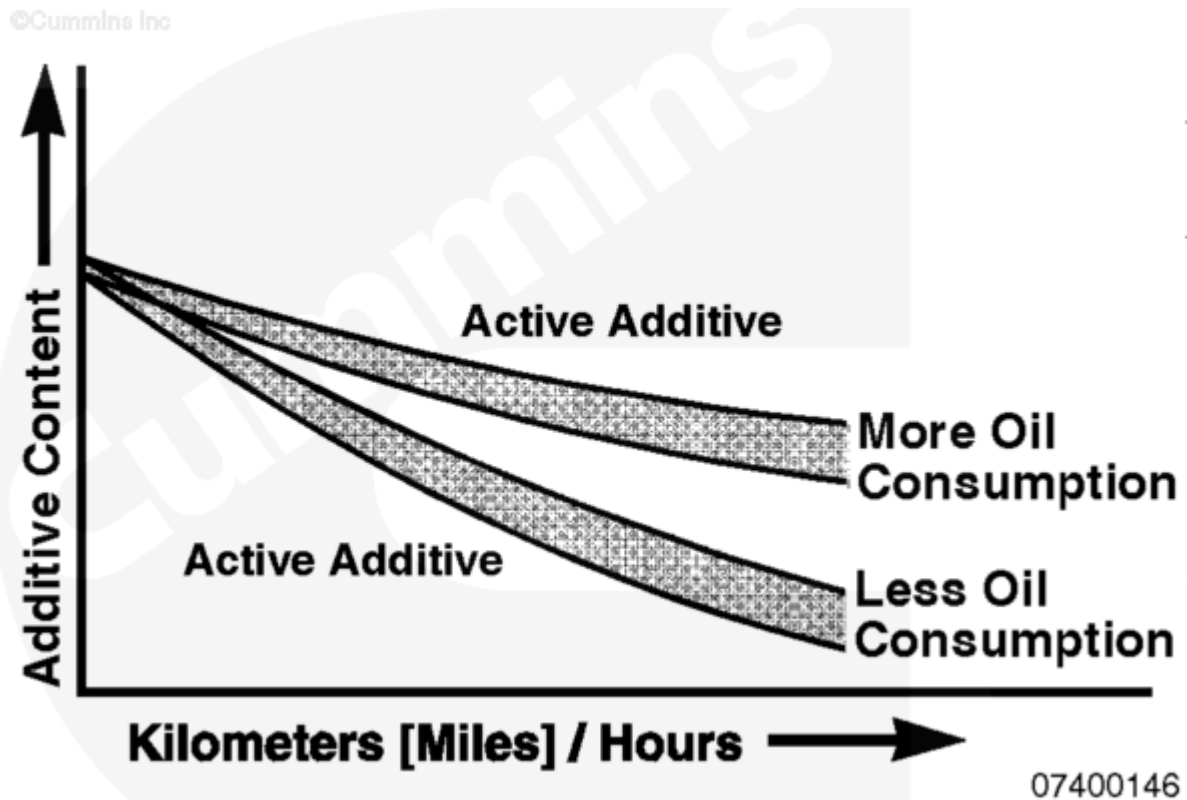
- Cooling System - Flush
- Fan Hub, Belt Driven - Check
- Fan Drive Idler Pulley Assembly - Check
- Water Pump - Inspect for Reuse
- Air Compressor Discharge Lines - Check
- Vibration Damper, Viscous - Check⁸

Maintenance Procedures at 10,000 Hours (Section 9)

- Fuel Injectors - Replace⁶
- Turbocharger - Check⁶

1. Under circumstances where hours of operation are **not** accumulated at a fast rate, use calendar time. In other words, use hours or calendar time, whichever comes first.
2. Cummins Inc. recommends the use of dry-type air cleaners.
3. Alternative oil drain information is available in the oil drain section of this procedure.
4. Mechanically actuated injector engines **only**.
5. Electronically actuated injector engines **only**.
6. Recommended at engine half life to rebuild. Half life varies by application. Contact a Cummins® Authorized Repair Location if the interval can **not** be determined. Alternative injector replacement information is available in the injector replacement interval section of this procedure.
7. **Only** when coupled with Chart Method explained in the oil drain section of this procedure.
8. Vibration dampers have limited service life. The damper **must** be replaced after 576,000 km [360,000 miles], or 15,000 hours of service.

Oil Drain and Fuel Filter Change Intervals



As the engine oil becomes contaminated, essential oil additives are depleted. Lubricating oils protect the engine as long as these additives are functioning properly. Progressive contamination of the oil between oil and filter change intervals is normal. The amount of contamination will vary, depending on the operation of the engine, kilometers, miles, or hours on the oil, fuel consumed, and new oil added.

Extending oil and filter change intervals beyond recommendations will decrease engine life, due to factors such as corrosion, deposits, and accelerated wear.

Use the oil drain interval recommendations in this section to determine which oil drain interval method to use for your application.

Do **not** extend oil and filter change intervals beyond 500 hours or 6 months (except for standby generator drive engines) unless the Chart Method is used. On standby generator drive engines, the intervals are 500 hours or 12 months, whichever occurs first. See the charts below.

There are three recommended methods to determine proper oil and filter change interval:

- Fixed Mileage Method (based on fixed kilometer, miles, hours, or months; whichever occurs first). This information is contained in the General Information section of this procedure.
- Centinel™ System, coupled with an oil analysis program.
- Chart method (based on known fuel consumption rates, coupled with oil analysis program).

- Refer to Cummins® Engine Oil and Oil Analysis Recommendations, Bulletin 3810340 (/qs3/pubsys2/xml/en/bulletin/3810340.html) for details on oil analysis for all three methods.

Centinel™ - With a continuously operating Centinel™ System, the oil drain interval can be extended until the oil analysis requires the oil to be changed or the oil is known to be contaminated. Oil analysis is required at 250 hour intervals when using the Centinel™ System, and **must** include soot measurement. Fleetguard® Oil Analysis Kit, Part Number CC 2543, meets this requirement. Oil analysis parameters **must** remain within the limits specified in Table 2 of the Oil Analysis Techniques for High Horsepower Diesel Engines, Service Bulletin 4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html). Reference the following bulletins for more information on oil sampling and analysis:

- Fluids for Cummins® Products Service Manual, Bulletin 5411406 (/qs3/pubsys2/xml/en/bulletin/5411406.html), Section 4 Engine Oil.
- Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html).
- Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452 (/qs3/pubsys2/xml/en/bulletin/2883452.html).

Premium oil filters, such as Fleetguard® LF 9000 series filters, or equivalent, are also required when using Centinel™ with spin-on filters.

Chart Method

The information listed below is required when using the chart method to determine the correct oil and filter change interval for the engine.

- Oil system capacity (sump plus any remote tank volume)
- Average fuel consumption rate
- Use engine oil that complied to the standards listed in Cummins® Engine Oil and Oil Analysis Recommendations, Bulletin 3810340 (/qs3/pubsys2/xml/en/bulletin/3810340.html).
- Oil analysis to demonstrate the chart method interval is acceptable. Reference the following service bulletins:
 - Cummins® Engine Oil and Oil Analysis Recommendations, Bulletin 3810340 (/qs3/pubsys2/xml/en/bulletin/3810340.html)
 - Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html)
 - Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452. (/qs3/pubsys2/xml/en/bulletin/2883452..html)

System capacity can be determined by knowing the volume of the oil required to touch the high-level mark on the dipstick and volume of any remote oil tanks on the machine in which oil is continuously circulated. Oil sump capacities are listed in the operation and maintenance manuals for all Cummins® engines. It is imperative that the total system volume be known when

using the chart method. Use the following procedure for information on the oil sump capacity. Refer to Procedure 018-017 in Section V (/qs3/pubsys2/xml/en/procedures/20/20-018-017.html) for system capacity. If the machine is equipped with an oil reserve system with a reservoir remote from the engine oil sump, the reservoir volume **must** be added to the engine sump volume to determine the total system capacity. This is **only** true for remote tanks in which the oil is continuously circulated.

When oil is changed on an engine with a remote oil tank in which oil is continuously circulated, the oil in the tank **must** be changed in addition to the oil in the engine sump.

To use the chart method effectively, accurate fuel consumption records **must** be kept and maintained. Fuel consumption information **must** be in liters per hour or gallons per hour units to use the charts. Fuel consumption rates can change over time because of an increasing or decreasing engine duty cycle. Accurate records are essential in determining average fuel consumption during a given oil change interval.

The charts are categorized and labeled by the system capacity for a specific engine family. Select the chart representing the correct oil system capacity for engine model in question. The left vertical axis of the chart represents fuel consumption in U.S. gallons per hour. The chart method shows both gallons per hour and liters per hour. Determine the intersection point of the fuel consumption rate on the applicable interval curve by drawing a horizontal line. From this point, draw a vertical line down until it intersects with the horizontal axis representing hours. This point represents the acceptable oil change interval.

Note : Before using the charts below, it is important to note the grade of oil being used. Reference Table 1. This can affect which chart can be used.

Table 1: Engine Oil Grades

North America Classification	International Classification	Cummins Engineering Standard (CES)	Description
API ¹ CD API CE	ACEA ² E1	Standard obsolete. These oils are no longer recommend for use in Cummins® engines	Do not use the Chart Method if using these oils.
API ¹ CG-4/SH			
API CF-4/SG	ACEA ² E2	CES 200757	Engines using these oil grades can use the Chart Method
	ACEA ² E3		
API ¹ CH-4 /SJ	JAMA ³ DH-1 ACEA ² E55	CES 20071	
		CES 20076 CES 20077	
API ¹ CI-4	ACEA ² E7	CES 20078	
API ¹ CJ-48	ACEA ² E9 JAMA3 DH-2	CES 20081	

Table 1: Engine Oil Grades

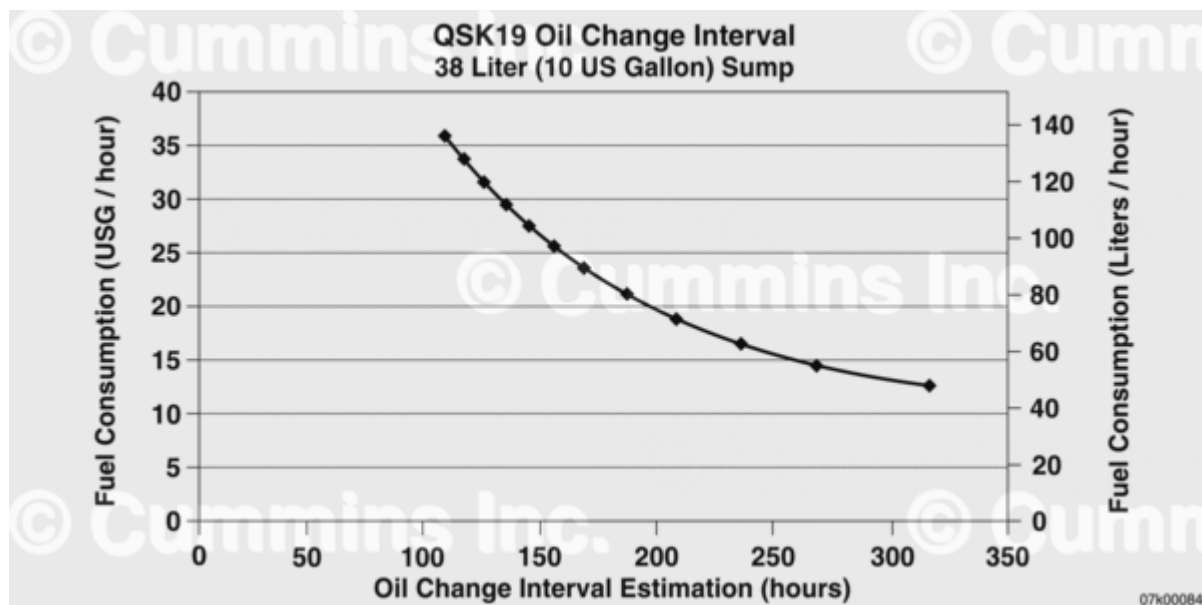
North America Classification	International Classification	Cummins Engineering Standard (CES)	Description
1. API - American Petroleum Institute. 2. ACEA - Association des Constructeurs European d' Association. 3. JAMA - Japan Automobile Manufacturer Association. 4. CES 20076 adds the requirement of a 300 hour Cummins® M11 test to API CH-4. 5. CES 20077 adds the requirement of a 300 hour test to ACEA ES. 6. Use of oils with only these designations poses an undue risk of engine damage for engines design to use more advanced oils. Even when drastically shortened oil change intervals are followed 7. CES 20075 CF-4/SH and E-3 oils can be used in areas where none of the recommended oils are available, but the oil drain interval must be reduced. Reference the appropriate maintenance schedule. Use of oils with only these designations poses an undue risk of engine damage for engines designed to use more advanced oils, even when drastically shortened oil change intervals are followed. 8. API CJ-4 oil should only be used on engines running with Ultra Low Sulphur Diesel fuel.			

⚠ CAUTION ⚠

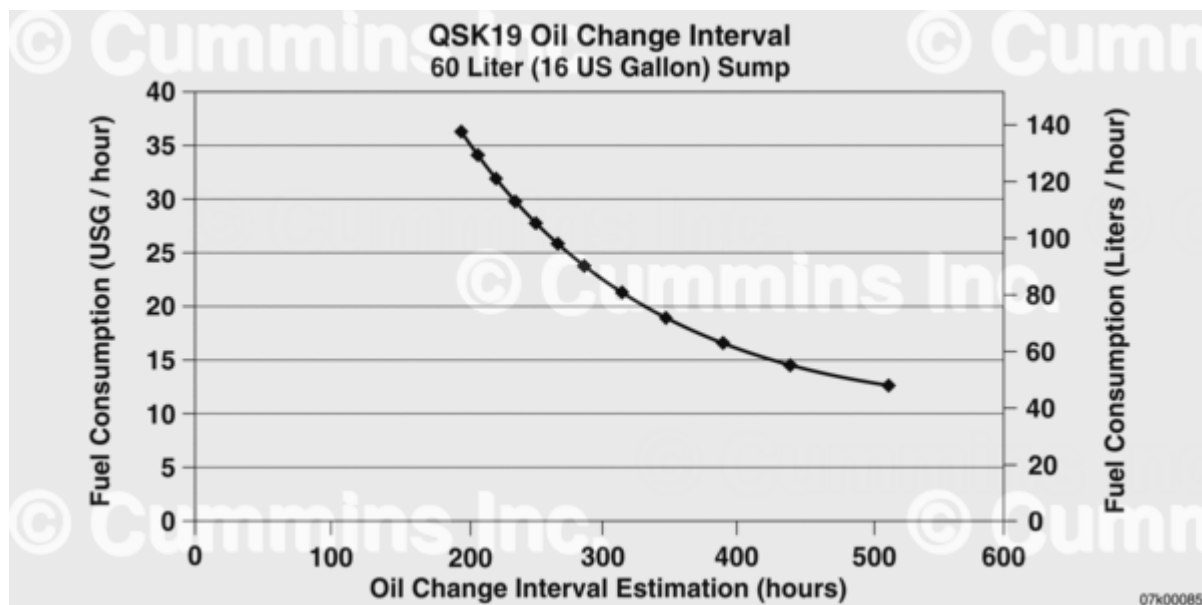
Inaccurate fuel consumption records can result in incorrect oil change intervals that can result in premature engine wear and/or deposits.

⚠ CAUTION ⚠

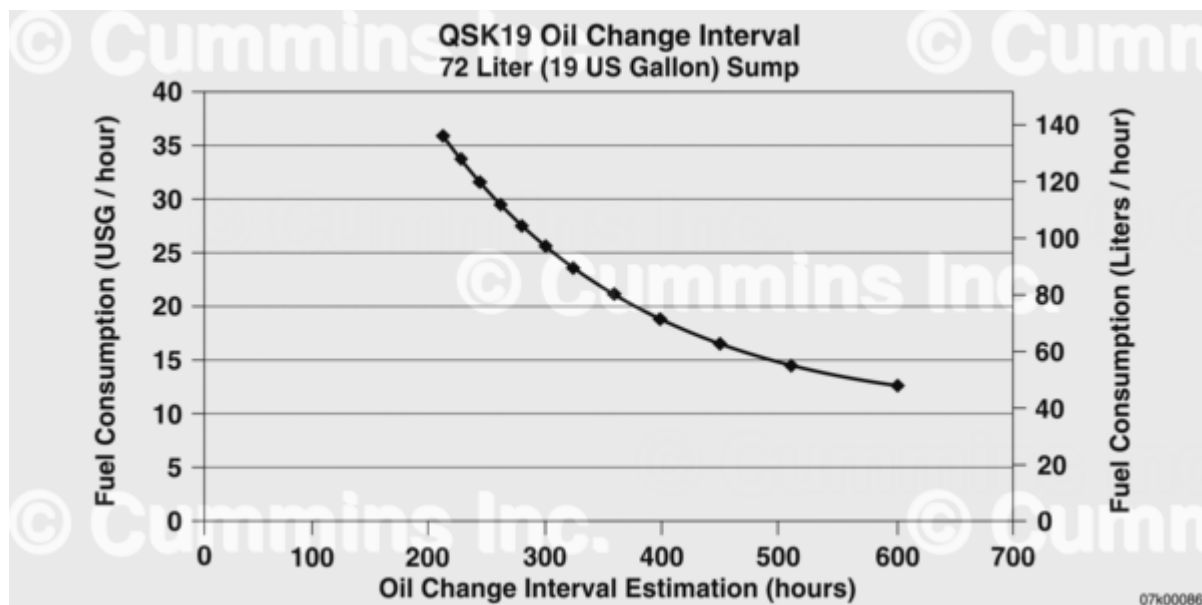
Do not extend oil change intervals beyond the maximum point shown when fuel consumption rates are less than the minimum shown. Doing so can result in shortened engine life due to oil degradation processes associated with lightly loaded engines



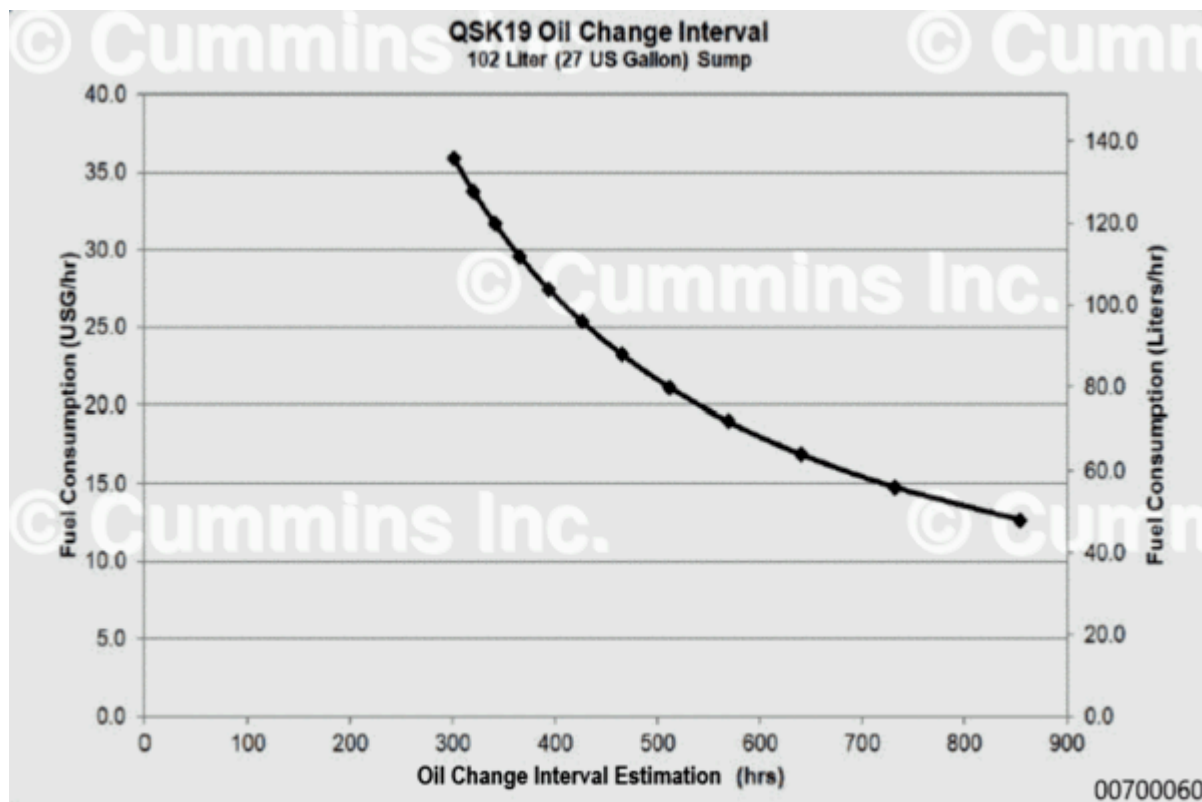
38 Liter (10 US Gallon) Sump



60 Liter (16 US Gallon) Sump



72 Liter (19 US Gallon) Sump



102 Liter (27 US Gallon) Sump

To read the chart:

Assume the engine has a 10 gallon sump capacity and the engine consumes 15 gallons of fuel per hour.

- Select the chart titled 38 liter (10 US Gallons).
- The left vertical axis of the chart represents fuel consumption in gallons per hour.

- Determine the location of 15 gallons on the left vertical axis and draw a horizontal line across the chart until it intersects the curve.
- At the point of intersection, draw a straight vertical line down to the bottom of the graph.
- The point at which the vertical line intersects the bottom horizontal axis of the graph indicates the oil change interval (approximately 275 hours in this example).

Fuel Filter Change Intervals for Electronically Actuated Injector Engines Only

- Fuel filter change intervals are set at 250 hours in order to protect the fuel system from hard particles.
- **Not** all fuel filters available on the market are equivalent, and may **not** protect the fuel system adequately beyond the 250 hour threshold.
- Cummins Inc. requires the following hardware to achieve Stage 1 and Stage 2 fuel filter change intervals beyond 250 hours, and up to but **not** exceeding 500 hours:
 - FH239 series Industrial Pro or FH240 series Sea PRO® filter head **must** be installed as the Stage 1 fuel filter head.
 - Fleetguard® NanoNet™ media fuel filters **must** be used on Stage 1 and Stage 2 fuel filters.
 - Use the following procedure for fuel filter part numbers. Refer to Procedure 018-024 in Section V. (</qs3/pubsys2/xml/en/procedures/20/20-018-024.html>)

Cummins Inc. does **not** recommend fuel filter change intervals exceed 500 hours.

The most common cause of degraded engine performance is poor fuel quality. Fleetguard® NanoNet™ media filters are designed to function with greater filtration efficiency than other filters, and thus retain more contaminants. If premature filter plugging is observed, additional filtration or improved fuel quality may be necessary to reach desired maintenance intervals. For more information on fuel cleanliness recommendations, reference the "Fuel Cleanliness" section of the Fuels For Cummins® Engines, Bulletin 3379001 (</qs3/pubsys2/xml/en/bulletin/3379001.html>).

Injector Replacement Interval

Injector replacement intervals are recommended at engine half life to rebuild for some injectors part numbers. Contact a Cummins® Distributor if **not** sure of this interval.

For some injector part numbers, if additional requirements are met, the injector replacement interval is at rebuild.

Injector Replacement Interval		
Injector Part Number	Description	Replacement Interval
4924587, 4928053, 4928106	C1 Injectors	Half life to rebuild
2881083, 4964170, 4964171	C2 Injectors	Half life to rebuild

Injector Replacement Interval		
Injector Part Number	Description	Replacement Interval
2881088	C3 Injectors without flow limiters	Half life to rebuild
3867149	C3 Injectors for T3 engines with unique nozzle	Half life to rebuild
2867147	C3 Injectors with flow limiters, not using NanoNet™ fuel filters ¹ or with fuel sulfur levels > 500 ppm	Half life to rebuild
2867147	C3 Injectors with flow limiters, using NanoNet™ fuel filters ¹ , fuel sulfur levels <500 ppm	Rebuild

¹Nanonet Fuel Filtration:

- FH239 series Industrial Pro or FH240 series Sea Pro® filter head **must** be installed as the Stage 1 fuel filter head.
- Fleetguard® NanoNet™ media fuel filters **must** be used on Stage 1 and Stage 2 fuel filters.
- Use the following procedure for fuel filter part numbers. Refer to Procedure 018-024 in Section V.

Coolant Drain Interval

Maintenance Intervals for Cooling System up to 568 liters [150 gal]					
System Size in liters [gal]	79-144 [21-30]	117-189 [31-50]	193-284 [51-75]	288-378 [76-100]	382-568 [101-150]
Hours	Units of SCA				
751-1000	25	50	80	75	150
501-750	20	35	60	100	110
251-500	15	25	40	50	75
0-250	10	15	20	25	40

Maintenance Intervals for Cooling System up to 1514 liters [400 gal]					
System Size in liters [gal]	572-757 [151-200]	761-946 [201-250]	950-1135 [251-300]	1139-1325 [301-350]	1329-1514 [351-400]
Hours	Units of SCA				
751-1000	200	250	300	350	400
751-1000	150	190	225	260	300
251-500	100	125	150	175	200
0-250	50	65	75	90	100

Note : Consult the vehicle equipment manufacturer's maintenance information for total cooling system capacity.

Note : When draining and replacing the coolant, always pre-charge the cooling system to a SCA level of 1.5 units per gallon. This concentration level must not be allowed to go below 1.2 units and must be controlled when the level is greater than 3 units. Action needed when the level goes below 1.2 is a filter and liquid pre-charge; from 1.2 to 3.0 units, filter only; above 3.0, test at every oil change until level falls to 3.0 or below.

When performing service which requires draining the cooling system, take special precautions to collect it in a clean container, seal it to prevent contamination, and save for reuse.

- A. Change coolant filters at each oil change to protect the cooling system. Consult the coolant capacity chart to determine the correct coolant filter for a given cooling system capacity and oil drain interval.
- B. Once the required number of SCA units has been determined for a given cooling system capacity and maintenance interval from the chart above, the table in the following procedure can be used to determine the correct filter part number. Refer to Procedure 018-024 in Section V. (</qs3/pubsys2/xml/en/procedures/20/20-018-024-om.html>)